GRAVEATLAS

Phase 2 — Core Data, Search, Map & Public Discovery

Full copy-paste implementation prompt for the Base44 agent.

COPY-PASTE MASTER PROMPT

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GRAVEATLAS — PHASE 2
CORE DATA, SEARCH, MAP & PUBLIC DISCOVERY

Phase 1 must already be complete.

Continue from the existing GraveAtlas implementation.

IMPORTANT:
- Do NOT rebuild Phase 1.
- Reuse the existing Android app, Cloudflare Worker/API, GitHub App, data schemas, authentication and repository structure.
- Do NOT start Phase 3.
- Do NOT add paid services without explicit approval.
- Do NOT fabricate cemetery, grave, person or historical information.
- Preserve provenance and data integrity.
- Do NOT expose private credentials or moderation data.
- Complete only Phase 2, then stop.

==================================================
PART 1 — CORE PUBLIC DATA MODEL
==================================================

Review and finalize the existing public schemas for:

- cemetery
- section/area
- grave/record
- person
- location
- source
- provenance
- media reference where supported

Every public factual record must have a traceable source/provenance field where applicable.

Do not invent missing values.

==================================================
PART 2 — DATA VALIDATION
==================================================

Implement schema validation for public data.

Validate:

- required fields
- field types
- dates
- coordinates
- identifiers
- references
- duplicate identifiers
- malformed records

Reject invalid data safely.

Do not silently modify historical facts.

==================================================
PART 3 — PUBLIC DATA INGESTION
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Create the foundation for loading approved public data from the GitHub data repository.

The backend must:

- read only the intended public dataset
- validate data
- handle malformed files safely
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- return controlled errors
- avoid exposing repository credentials

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PART 4 — CEMETERY SEARCH
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Implement public cemetery search.

Support appropriate search dimensions such as:

- cemetery name
- location
- area
- country/region where applicable

Search must be:

- validated
- paginated
- bounded
- safe against excessive queries

Do not return unlimited datasets.

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PART 5 — RECORD SEARCH
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Implement public record/person search where supported.

Support appropriate fields such as:

- name
- cemetery
- section
- date information
- record identifier

Do not expose private moderation or contributor information.

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PART 6 — SEARCH NORMALIZATION
==

Normalize search input safely.

Consider:

- whitespace
- case
- Unicode
- punctuation
- common formatting differences

Do not alter authoritative stored data merely for search normalization.

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PART 7 — SEARCH PAGINATION
==

Implement bounded pagination.

Require safe limits for:

- page size
- offset/cursor
- result count

Prevent requests from retrieving enormous datasets.

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PART 8 — SEARCH RESULT STRUCTURE
==

Define consistent result responses.

Each result should provide enough information for the user to understand the result and navigate to the full record.

Do not expose internal database implementation details.

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PART 9 — CEMETERY DETAIL
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Create the public cemetery detail experience.

Show applicable:

- cemetery name
- location
- description
- coordinates
- available record count where reliable
- sources
- relevant public metadata

Do not invent descriptions or counts.

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PART 10 — RECORD DETAIL
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Create the public record detail experience.

Show applicable:

- person/record name
- cemetery
- location
- dates
- source/provenance
- public notes
- correction/report entry point

Do not expose contributor-private information.

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PART 11 — MAP FOUNDATION
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Create the map foundation for cemetery/record locations.

Use coordinates only when available and validated.

Do not fabricate coordinates.

Handle invalid or missing coordinates gracefully.

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PART 12 — MAP MARKERS
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Implement appropriate public map markers.

Avoid loading every marker in the entire dataset at once.

Use:

- bounded queries
- geographic filtering
- clustering or equivalent where practical

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PART 13 — MAP SEARCH
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Allow the user to search or navigate the map where supported.

Prevent unnecessarily large geographic queries.

Return controlled results.

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PART 14 — LOCATION DETAILS
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When a cemetery has valid coordinates, allow the user to view its location.

If external navigation is supported, use a safe intent/link mechanism.

Do not hard-code a single map provider unless already required by the architecture.

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PART 15 — API ROUTES
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Implement/review appropriate public routes:

GET /health
GET /cemeteries
GET /cemeteries/{id}
GET /records
GET /records/{id}
GET /search
GET /map

Use the project's actual API conventions if different.

Do not expose private/admin routes as public routes.

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PART 16 — API VALIDATION
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Validate:

- path parameters
- query parameters
- page sizes
- search strings
- geographic bounds
- identifiers

Return consistent validation errors.

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PART 17 — API CACHING
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Use the Phase 1/architecture caching foundation where appropriate.

Only cache safe public responses.

Never cache:

- private user information
- admin responses
- moderation notes
- authentication credentials

Respect cache invalidation when public data changes.

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PART 18 — ANDROID PUBLIC DISCOVERY
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Implement the main public discovery flow.

User should be able to:

1. open GraveAtlas
2. search
3. see results
4. open cemetery/record
5. view location where available
6. return to results

Maintain loading, empty, offline and error states.

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PART 19 — SEARCH UI
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Create a usable search interface.

Include:

- search field
- search action
- result list

- pagination/incremental loading
- empty state
- error state
- retry

Do not overcomplicate the UI.

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PART 20 — CEMETERY UI
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Create the cemetery detail screen.

Include applicable public information.

Clearly distinguish:

- verified/available information
- source information
- unavailable information

Do not imply verification that does not exist.

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PART 21 — RECORD UI
==

Create the record detail screen.

Include:

- core public information
- source/provenance
- location if available
- report/correction entry point

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PART 22 — MAP UI
==

Create the public map experience.

Support:

- markers
- selection
- basic navigation
- loading state
- empty state
- unavailable location state

Do not require unnecessary location permissions just to browse public cemetery coordinates.

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PART 23 — DATA QUALITY
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Add safe checks for:

- duplicate IDs
- missing names
- invalid coordinates
- impossible date ranges
- broken references
- malformed source records

Flag questionable records rather than deleting them automatically.

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PART 24 — PERFORMANCE
==

Optimize public discovery for reasonable performance.

Use:

- pagination
- bounded queries
- caching
- lazy loading

- appropriate image handling if applicable

Do not add paid infrastructure.

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PART 25 — SECURITY
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Verify:

- public routes expose only public data
- admin/moderation data remains protected
- repository credentials stay server-side
- input is validated
- rate-limit hooks remain available
- errors do not expose secrets

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PART 26 — TESTING
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Create/run tests for:

- cemetery search
- record search
- pagination
- invalid search
- cemetery detail
- record detail
- map queries
- invalid coordinates
- malformed records
- API authorization boundaries
- caching behavior where testable

Do not invent test results.

==
PART 27 — DOCUMENTATION
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Create/update:

docs/PUBLIC-DATA.md
docs/SEARCH.md
docs/MAP.md
docs/API-PUBLIC.md
docs/DATA-VALIDATION.md

Document actual behavior only.

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PHASE 2 ACCEPTANCE GATE
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Verify:

[ ] Public data schemas
[ ] Data validation
[ ] Public data ingestion
[ ] Cemetery search
[ ] Record search
[ ] Search normalization
[ ] Pagination
[ ] Search result structure
[ ] Cemetery detail
[ ] Record detail
[ ] Map foundation
[ ] Map markers
[ ] Map search
[ ] Location handling
[ ] Public API routes
[ ] API validation
[ ] Public caching
[ ] Android discovery
[ ] Search UI
[ ] Cemetery UI
[ ] Record UI
[ ] Map UI
[ ] Data-quality checks

[ ] Performance
[ ] Security
[ ] Tests
[ ] Documentation

==
FINAL REPORT
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Return:

PHASE 2 STATUS:
READY / NOT READY

DATA MODEL:
PASS / FAIL

VALIDATION:
PASS / FAIL

SEARCH:
PASS / FAIL

CEMETERY DETAIL:
PASS / FAIL

RECORD DETAIL:
PASS / FAIL

MAP:
PASS / FAIL

API:
PASS / FAIL

ANDROID:
PASS / FAIL

DATA QUALITY:
PASS / FAIL

PERFORMANCE:
PASS / FAIL

SECURITY:
PASS / FAIL

TESTS:
PASS / FAIL

DOCUMENTATION:
PASS / FAIL

==
ACTUAL TEST RESULTS
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Report actual results only.

Cemetery search:
[actual result]

Record search:
[actual result]

Pagination:
[actual result]

Cemetery detail:
[actual result]

Record detail:
[actual result]

Map:
[actual result]

Invalid data:
[actual result]

Security:
[actual result]

Do NOT invent results.

==
BLOCKERS
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List every issue preventing:

PHASE 2 = READY

For each:
- severity
- component
- problem
- impact
- recommended remediation

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NON-BLOCKING ISSUES
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List remaining improvements.

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STOP CONDITION
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Do NOT start Phase 3.

Do NOT add paid services without explicit approval.

Do NOT fabricate data or test results.

Do NOT expose secrets.

Do NOT delete legitimate cemetery or historical data.

STOP after the final report.

WAIT FOR EXPLICIT INSTRUCTIONS FOR PHASE 3.